Seyed Ali Rashidi

Computer Vision & ML Engineer

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Profile

Computer Vision & ML Engineer with a track record building end-to-end deep learning pipelines for scientific imaging — spanning medical CT/MRI segmentation, astronomical image classification, and spaceborne thermal-infrared systems. Currently Payload Supervisor on the IGNIS CubeSat mission at the University of Naples Federico II, leading SNR modelling, Zemax optical simulation, and imaging performance analysis for a thermal-IR Earth observation payload. Proficient in PyTorch, TensorFlow, OpenCV, and Docker, with production-deployed models on Google Cloud Run. Brings cross-domain depth across computer vision, sensor physics, and containerised ML systems.

Skills

Languages & Frameworks — Python, C++, C, Bash, PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Streamlit

Computer Vision & ML — Image classification, object detection, semantic & instance segmentation, CNNs, ResNet, U-Net, YOLOv8, transfer learning, Optuna, data augmentation, class imbalance handling, model evaluation (mAP, IoU, Dice, precision/recall)

Deployment & MLOps — Docker, REST inference APIs, Google Cloud Run, GCP, Git, CI/CD, model versioning, containerised deployment, Linux (Bash), Jupyter, MLflow

Imaging & Sensor Systems — Thermal IR imaging, radiometric modelling, SNR/NETD analysis, FLIR Boson 640, FoV estimation, Zemax OpticStudio, statistical learning, data mining, time-series analysis, anomaly detection

Professional Experience

03/2025 – Present

University of Naples "Federico II" - IGNIS CubeSat Mission
Payload Supervisor — Computer Vision & Imaging Systems

- Built physics-aware SNR and imaging performance models for a thermal-IR CubeSat payload using FLIR Boson 640 data; simulated full optical system in Zemax OpticStudio — modelling FoV, spatial resolution, and ground footprint across orbital scenarios.
- Identified non-optimal launch timing through time-series SNR analysis; recommended a ~5-month delay, directly improving projected early-mission image quality.
- Conducted payload hazard analysis and supported engineering trade-offs with multidisciplinary spacecraft teams.

01/2016 – 12/2023

EZ-Tech — Freelance / Self-Employed
Data Recovery Engineer

- Operated an independent data recovery practice, diagnosing and recovering data from failed storage systems across hundreds of client cases using low-level signal integrity analysis and structured failure diagnosis.
- Managed client relationships, case workflows, and quality delivery independently under strict time and technical constraints.

Projects

04/2026 – Present

Brain CT Hemorrhage Segmentation 🔗
Intracranial Hemorrhage Detection — U-Net CT Segmentation

- Built an end-to-end segmentation pipeline for intracranial hemorrhage detection from CT scans using U-Net in PyTorch; implemented 3D→2D slicing, NIfTI processing, Hounsfield Unit windowing, and balanced patch sampling for class imbalance.
- Custom loss combining BCEWithLogitsLoss and Dice Loss; deployed as a Dockerized REST API on Google Cloud Run with Streamlit inference interface.

08/2024 – 11/2024

Radio Galaxy Classification 🔗
Morphological Classification of Extragalactic Radio Sources — CNN

- Designed an end-to-end CV pipeline classifying radio galaxies (FR0, FRI, FRII) from ~18,000 grayscale images (50×50px) using a custom CNN in TensorFlow; plain CNN achieved 82.77% test accuracy, outperforming ResNet (77.81%) and Optuna-tuned variants (80.28%).

- Deployed as a Dockerized REST inference API on Google Cloud Run — production-ready, scalable HTTP-based inference.

07/2025 – 11/2025 **Thermal Infrared Heatmap & SNR Prediction**
- Built a Python-based physics-aware simulation framework predicting SNR and imaging performance for a thermal-IR CubeSat payload; modelled signal levels as a function of orbital altitude, viewing geometry, and temperature contrast.
- Generated thermal heatmaps and conducted sensitivity analysis on ground footprint and observation geometry; outputs directly informed IGNIS payload design decisions.

02/2026 – Present **Brain Tumor Segmentation — YOLOv8 on Multimodal CT/MRI**
- Training a YOLOv8-seg model for automated brain tumor segmentation on multimodal CT and MRI datasets using transfer learning in a GPU-accelerated environment.
- Evaluating with box/mask mAP, precision, recall, and Dice score; applying augmentation strategies for cross-modality robustness.

03/2025 – 05/2025 **Reinforcement Learning for Forex Trading**
- Built a DQN-based RL agent in PyTorch for EUR/USD trading using OHLC time-series data; implemented experience replay buffer and epsilon-greedy exploration with independent action, stop-loss, and target selection (target = 2× SL).
- Applied maximum holding period constraints and reward shaping to simulate realistic trading conditions; ongoing experimentation with state representation and policy refinement.

Education

09/2023 – present
Naples, Italy
University of Naples Federico II
Master’s Degree in Data Science
Thesis: Signal-to-Noise Ratio Characterisation for Thermal-Infrared Microbolometric Imaging in CubeSat Missions
Key courses: Machine Learning, Computer Vision, Statistical Learning, Data Mining, Big Data, Advanced Statistical Modelling

09/2009 – 01/2012
Islamic Azad University
BSc in Computer Software Engineering
Data Structures, Algorithms, Systems Design, Programming, Engineering Mathematics

Languages

English	**Italian**	**Persian**
B2	A2	Native

I hereby give consent for my personal data to be processed for recruitment purposes in accordance with EU Regulation 2016/679 (GDPR).